

Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

RP 25-26J-264

Project Proposal Report

S.W.O.P.K. WADUGE

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology
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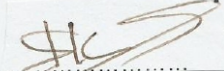
DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
S W O P K WADUGE	IT22586384	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



Signature of the supervisor:

(Mr. Uditha Dharmakeerthi)

Date: 30/08/25

Abstract

Sri Lankan mushroom farming is becoming more and more well-known as a lucrative agribusiness, yet conventional methods for regulating substrate pH and moisture frequently provide less-than-ideal yields. This proposal provides an overview of the creation of a predictive advising system to help growers of mushrooms control the pH and moisture content of their growing media. In addition to machine learning models (LSTM/GRU and ARIMA) to predict future trends in moisture and pH, the system incorporates Internet of Things sensors to track ambient and substrate conditions and an advice engine to suggest prompt treatments. Iterative development of the system's design, which includes sensor nodes, an edge computing unit, cloud analytics, and a dashboard that faces farmers, will be guided by an Agile methodology. We go over the project's goals, methodology, system architecture, and assessment strategy in depth. In oyster mushroom farms, the suggested method seeks to improve productivity and quality by facilitating data-driven, accurate control of growing factors. In addition to a thorough literature evaluation of smart farming approaches, functional and non-functional requirements, a budget estimate, and future implementation and testing strategies, the report adheres to SLIIT formatting standards and IEEE citation style. According to the findings of related studies, predictive analytics and IoT-driven monitoring can greatly enhance agricultural results and resource efficiency. This initiative helps to promote sustainable and lucrative mushroom farming by providing Sri Lankan mushroom growers with an intelligent guidance tool.

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Introduction

Background & Literature Survey

Mushroom civilization has surfaced as a promising agribusiness in Sri Lanka, driven by growing consumer demand for nutritional, protein-rich food and by the capability to grow mushrooms in fairly small spaces [1]. Oyster mushrooms [*Pleurotus ostreatus*], in particular, are popular due to their culinary versatility and health benefits [1]. In recent times, the Sri Lankan mushroom sector has endured steady growth, reflecting global trends. It's projected that Sri Lanka's mushroom request will achieve a robust periodic growth rate of roughly 9.4 by 2027, which is on par with leading directors like China and India [8]. This rapid-fire growth underscores the significance of technological invention to ameliorate productivity and meet the rising request demand. As shown in Figure 1, Sri Lanka's mushroom sector is projected to grow at about 9.4% CAGR by 2027, a rate comparable to leading Asian markets [8].

**Mushroom Cultivation Market: Sri Lanka vs Top 5 Major Economies in 2027
(Asia)**

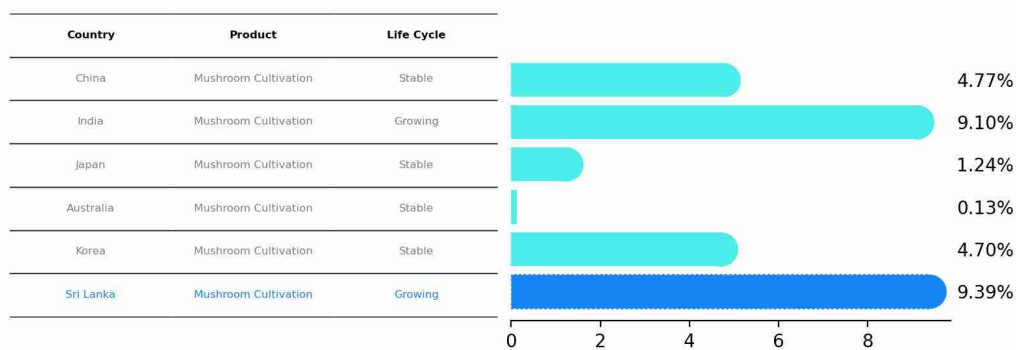


Figure 1. Projected mushroom cultivation market growth (CAGR by 2027) — Sri Lanka vs. top Asian economies.

In Sri Lanka, mushrooms are generally grown by small-scale growers, frequently in simple inner setups or “grow apartments” [1]. Maintaining optimal growing conditions in these surroundings is challenging due to the country's tropical climate and the perceptivity of mushrooms [9]. Crucial environmental factors similar as temperature, moisture, and light must be precisely controlled to insure healthy mycelium growth and fructification [9]. Key environmental drivers relevant to substrate health are summarized in Figure 2.

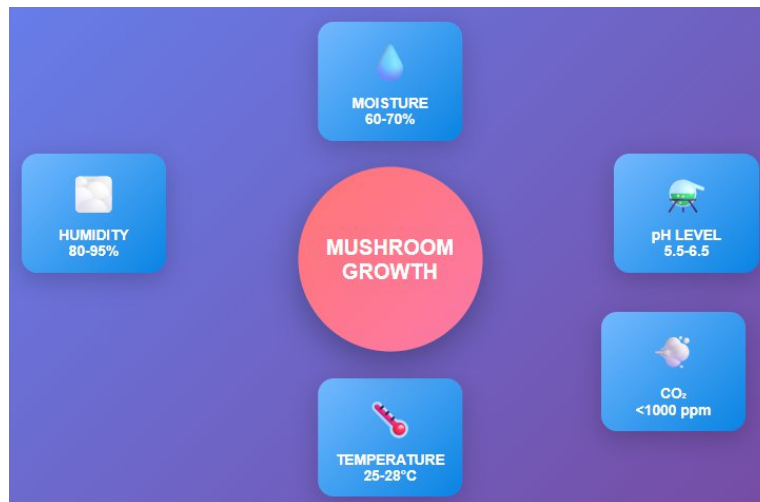


Figure 2. Key environmental factors affecting mushroom growth: moisture, pH, temperature, humidity, CO₂.

Among these factors, the humidity content of the civilization substrate and the substrate's pH position are critical parameters – yet these are generally overlooked in traditional husbandry practices [7]. Proper substrate humidity ensures that mushrooms admit acceptable water for development, and a stable pH in the growing medium is essential for nutrient vacuity and mycelial health [7]. Neglecting these substrate conditions can thus limit mushroom yield and quality. A photographic example of a typical small-scale oyster mushroom grows-room is shown in Figure 3.



Figure 3. Small-scale oyster mushroom farm in Sri Lanka — manual watering and simple facilities

Smart husbandry and perfection husbandry ways [the core of the “Agriculture 4.0” movement] have begun to access Sri Lanka’s agrarian sector. The Internet of effects [IoT] – a network of connected detectors and bias – has proven particularly salutary by enabling real-time monitoring and control of ranch conditions [3]. For illustration, IoT-based irrigation systems in analogous tropical climates [similar as in Malaysia] have achieved over 24 water savings compared to homemade styles [4]. Similar successes motivate the operation of IoT in mushroom civilization, where nonstop monitoring of the microclimate and substrate could significantly ameliorate yields and reduce labor conditions. By espousing IoT results, indeed small mushroom granges can be converted from static, manually managed operations into dynamic, data-driven systems that optimize growing conditions around the timepiece.

Research Gap

pH Monitoring and Vaticination

Mushrooms generally bear a slightly acidic substrate; for oyster mushrooms, an optimal pH is around 5.5 – 6.5 [7]. Over the course of the civilization cycle, the substrate pH can shift due to metabolic derivations of the fungi and other microorganisms, as well as due to the effects of repeated watering. Presently, no given original result provides nonstop pH monitoring of mushroom grow-beds. Research from other countries has begun to explore automated pH control; for case, Jenath Revanth et al. developed an IoT-grounded system that monitors soil pH along with temperature and moisture for oyster mushroom civilization and automatically adds chemicals [e.g., lime result] to neutralize the pH when it drifts out of range [2]. That system, still, was substantially a proof of concept and didn't incorporate any predictive analytics. Therefore, the linked exploration gap isn't only the capability to sense and adjust substrate pH in real-time, but also the capability to prognosticate when the pH is likely to drift out of the optimal range and to advise proactive measures before yields are affected.

Predictive Advisory vs. Robotization

Another gap lies in the decision-support approach for farm operation. Some advanced mushroom tilling systems apply full robotization – directly controlling actuators like fans, heaters, humidifiers, or pumps to adjust conditions in a closed-loop manner [7]. While full robotization can be effective in theory, in small-scale husbandry contexts numerous farmers prefer to retain control and have a predictive system rather than a completely automatic one, due to issues of trust, cost, and the desire to apply their own judgment. Many existing systems give a predictive advisory; for illustration, alerting a grower that “substrate humidity will probably drop below X in the coming 6 hours, consider watering soon,” or that “substrate pH is trending high; add an acidic supplement within 48 hours.” Our design explicitly aims to fill this gap by focusing on predictive analytics [using time-series forecasting models] coupled with real-time advisories. This approach leverages technology to enhance grower decision-making rather than replace it, which can ameliorate adoption rates among traditional growers who may be cautious of fully autonomous systems.

A review of existing results and literature reveals a clear gap in the sphere of mushroom civilization operation. While automated climate control systems for mushroom farms exist – for illustration, systems that regulate temperature and air moisture using sensors and actuator feedback loops [6], [9] – many results concentrate on substrate-specific parameters like the humidity content of the growing medium or its pH value. Most IoT-grounded mushroom husbandry systems to date have emphasized controlling ambient conditions. For example, numerous designs use DHT-type sensors to cover air temperature and relative moisture and also trigger fans, heaters, or humidifiers to maintain those ambient conditions [6], [9]. These ambient-control systems are important, but they don't directly address the conditions of the substrate where the mushrooms actually grow.

Substrate Humidity vs. Air Humidity

Traditional automated systems tend to maintain air moisture in the cultivation room [e.g., spraying mist when relative moisture falls below a threshold] [6]. Still, air moisture is only an indirect indicator of water available to the mushrooms. The substrate can gradually dry out even if the ambient moisture is kept high – especially toward the end of a growth cycle or if substrate preparation was inconsistent – because the water content in the substrate itself may be depleted. Conversely, the substrate can be excessively wet while air moisture readings appear normal. In practice, the humidity content of the substrate is a more direct index of water accessible to the mushroom mycelium. This approach addresses that gap by incorporating soil humidity sensors directly into the substrate bags to monitor water content in real-time, an approach infrequently seen in previous mushroom cultivation systems.

Lack of Local Data and Dashboards

A final gap identified is the absence of localized data collection and visualization tools for mushroom cultivation in Sri Lanka. At present, growers don't keep detailed records of how substrate humidity and pH levels vary or how those variations relate to crop yield and health. By developing a system that logs these environmental and substrate parameters along with cultivation activities, we contribute to creating a valuable dataset for future analysis. Additionally, the proposed grower-facing dashboard [see Figure 1.3 in our design] will display real-time sensor readings and model predictions in an intuitive manner [e.g., graphs of humidity levels over time, pH trend indicators, and alert notifications]. Presently, similar data-driven visualization isn't available to these growers. Addressing this information gap – essentially making invisible conditions visible – will enable growers to make more informed adjustments based on evidence rather than guesswork.

In summary, this exploration targets the niche of substrate-centric, predictive operation in mushroom cultivation. It extends beyond prior work by integrating substrate humidity sensing, nonstop pH monitoring, and predictive algorithms into a single advisory framework optimized for small-scale oyster mushroom farms in Sri Lanka. This focus directly addresses the identified gaps and aims to empower growers with practical insights that were previously unattainable in traditional mushroom husbandry practices.

Research A: Mohammed et al. (2018) – In one example study, researchers developed an IoT-based monitoring and environmental control system for indoor oyster mushroom cultivation. Their system used sensors to track conditions like temperature and humidity and automatically controlled devices (heaters, fans, etc.) to maintain those parameters. Limitation: This project focused on general climate control and did not incorporate substrate moisture sensing or any predictive analytics. Contribution: It demonstrated that IoT technology can successfully automate the ambient environment of a mushroom house, improving stability of temperature and humidity. How Our Project Differs: We extend beyond ambient conditions by directly monitoring the substrate's moisture and pH and by adding a predictive advisory component for proactive decision-making, rather than only reactively controlling the climate.

Research B: Surige et al. (2021) – This study designed an IoT-based monitoring system for oyster mushroom farming in Sri Lanka. The system collected data from sensors (likely ambient conditions) and could alert the farmer, serving as a tech-enabled monitoring tool. Limitation: The solution was limited to real-time monitoring and basic alerts; it did not provide recommendations or predictions, nor did it specifically target substrate moisture or pH management. Contribution: It provided a baseline for how local mushroom farmers can use IoT sensors and cloud connectivity to observe farm conditions remotely, indicating the feasibility of such technology in the Sri Lankan context. How Our Project Differs: Our system not only monitors conditions but also predicts future substrate states (moisture and pH) and gives actionable advice. We place sensors in the substrate itself (an area Research B did not emphasize) and use machine learning models to forecast and guide interventions, thus offering a more advanced decision support system.

Research C: Rebaka et al. (2023) – This recent work developed an IoT-based automation system for oyster mushroom cultivation, including control of temperature, humidity, and even substrate pH levels. Their setup featured sensors and actuators to automatically regulate the grow-room climate and periodically adjust substrate pH via an automated mechanism. Limitation: The system, while comprehensive in automation, operated on reactive control logic (e.g., turning devices on/off based on threshold breaches) and did not incorporate predictive modeling or forecasting. It lacked a decision-support interface for the farmer, since it aimed for full automation. Contribution: This study showed that an integrated sensor-actuator network can successfully maintain mushroom cultivation parameters (including substrate pH) within desired ranges, leading to more stable growing conditions. How Our Project Differs: Instead of full automation, our project emphasizes a predictive advisory approach: we forecast future conditions and advise the farmer, allowing them to take action proactively. This human-in-the-loop strategy is more cost-effective and user-friendly for small growers. Additionally, while Research C's system kept conditions in range through automation, it did not provide insight or data to the farmer; our system will log data and present it via a dashboard, empowering the farmer with knowledge about trends and system performance.

Application Reference	Real-time Environmental Monitoring (Moisture, pH)	Automated Environmental Control	Remote / Mobile Access	Predictive Analytics & Advisory	Fruiting Readiness Forecasting	Farmer Decision Support	Cost-effective, Small-scale Suitability
Research A (Mohammed et al., 2018)	✓	✓	X	X	X	X	X
Research B (Surige et al., 2021)	✓	✓	X	X	X	X	✓
Research C (Rebaka et al., 2023)	✓	✓	✓	✓	X	X	X
Proposed System	✓	✓	✓	X	X	✓	✓

Figure 4 illustrates the research gap between existing monitoring/control systems and the proposed substrate-centric predictive advisory approach.

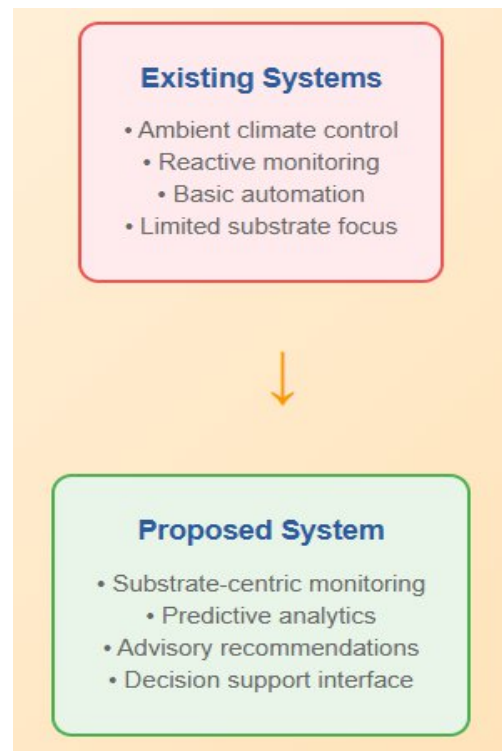


Figure 4. Research gap: comparison of existing ambient-control and monitoring systems versus the proposed predictive advisory system focused on substrate metrics.

Research Problem

Despite the clear growth potential of mushroom husbandry in Sri Lanka, most traditional growers still rely on conventional methods and often lack confidence in managing critical growing conditions. Traditional practices include watering mushroom bags on fixed schedules or based on visual cues and adjusting substrate pH periodically [e.g., by adding lime] without any factual measurements [5], [7]. These imprecise methods lead to several problems:

Inconsistent Substrate Humidity

Farmers generally spray or mist the mushroom cultivation bags to maintain moisture. Without sensor feedback, they have no accurate knowledge of actual substrate humidity levels, often resulting in over-watering or under-watering. Over-watering can promote bacterial or fungal growth, whereas under-watering stresses the mycelium and reduces yield [5].

Unmonitored pH Oscillations

The pH of mushroom substrates [generally composed of materials like sawdust, straw, or coir plus supplements] can drift over time due to microbial activity and repeated watering. Currently, growers adjust pH using a “schedule system” – for example, adding lime every couple of weeks – without measuring the current pH [2]. This crude approach can lead to unsuitable pH conditions that reduce nutrient uptake and inhibit mushroom growth.

Labor-Intensive Monitoring

Maintaining proper environmental conditions [moisture, temperature, humidity] requires constant attention. Growers must manually check growing areas and perform tasks like misting multiple times a day, which is labor-intensive and prone to human error or oversight [e.g., missing a scheduled misting], especially for smallholder growers who may have other jobs or responsibilities [1]. This continual attention increases labor costs and grower stress.

Yield Loss and Quality Issues

As a consequence of these challenges, mushroom yields under conventional practices frequently remain below their optimal potential. When humidity and pH diverge from ideal ranges, mushrooms can take longer to develop, have lower weight, or become more susceptible to contamination and spoilage [1]. Farmers therefore face reduced profits and inconsistent crop outcomes under the status quo.

For many smallholder farmers – including women and young entrepreneurs who have turned to mushroom cultivation as a source of income – these challenges pose a threat to sustainable operations. A single “bad batch” caused by improper humidity or pH can result in significant financial loss. Ensuring stability and reliability in product quality is thus a socio-economic imperative.

The core problem can be summarized as follows: **How can we provide Sri Lankan mushroom growers with a smart, predictive advisory system that helps maintain substrate humidity and pH at optimal levels, thereby maximizing yield and reducing labor?**

Objectives

Main Objectives

To design and implement a predictive mushroom fruiting system that combines real-time IoT environmental sensing with machine learning to forecast fruiting readiness, thereby improving yield consistency, sustainability, and decision-making for small-scale mushroom farmers in Sri Lanka [3][4][5].

Specific Objectives

To achieve the main objective, we outline several Specific, Measurable, Achievable, Relevant, and Timebound (SMART) objectives:

Objective 1: Sensor Integration and Monitoring

- Implement an IoT sensor network to monitor substrate moisture, pH, and ambient temperature/humidity.
- Target $\pm 5\%$ accuracy for moisture and ± 0.2 for Ph.
- Complete hardware setup and calibration by Month 3.

Objective 2: Data Logging and Dashboard Development

- Stream sensor data to a cloud server and store historical data.
- Develop a dashboard showing real-time and historical readings, accessible 24/7.
- Deploy a functional prototype by Month 4.

Objective 3: Predictive Modeling

- Use machine learning models [LSTM/GRU, ARIMA] to forecast moisture (6–12 hours) and pH (24 hours).
- Target RMSE $< 5\%$ for moisture and < 0.3 pH units, with MAE within acceptable bounds.
- Complete model development and validation by Month 6.

Objective 4: Advisory Engine and Notification System

- Generate predictive alerts based on sensor data and model forecasts.
- Notify farmers via dashboard, SMS, or mobile app; ensure $\geq 90\%$ accuracy.
- Complete advisory logic and notification integration by Month 7.

Objective 5: System Testing and Evaluation

- Test the system in lab and small-scale farm settings; evaluate yield, water usage, and labor reduction.
- Target $\geq 15\%$ yield improvement or 20% water savings.
- Complete testing and gather results by Month 8; finalize report by Month 9.

Methodology

Overview

This section describes the methodological framework used to design, implement, and validate a predictive advisory system for mushroom cultivation. The principal objective is to deliver a working prototype that combines reliable sensor data acquisition, robust short-horizon forecasting for substrate moisture and pH, and an advisory engine that translates predictions into clear recommendations for growers. The approach balances engineering rigor and empirical science: engineering to produce a modular, testable system; experimentation to generate labelled data for model training and evaluation; and iterative validation with domain experts to ensure practical relevance and usability. Key assumptions include limited initial datasets, preference for low-cost, field-robust hardware, and intermittent network connectivity in target deployments; these assumptions guide sensor choice, edge buffering, and model selection.

Research design and workflow

The project adopts an applied, iterative development strategy: build, test, collect data, refine. Work is organized into short sprints (typically 2–3 weeks) to enable rapid feedback loops and to contain research spikes—time-boxed investigations into technical unknowns such as pH interfacing or sensor conditioning. Each sprint concludes with a review involving the supervisory team and a cultivation domain expert to validate assumptions, inspect results, and prioritize corrective actions. This workflow is chosen to manage hardware variability and environmental unpredictability while maintaining steady progress toward deliverables [1],[9].

System architecture

(layered IoT design) The system follows a four-layer, modular IoT architecture: sensor (data acquisition), edge (local processing and buffering), cloud/backend (storage, analytics, advisory engine), and application (dashboard and notifications). This separation of concerns supports independent testing and staged deployment and enables the system to evolve from advisory-only to optional actuator integration in future work [1],[3]. Figure 5 presents the development cycle and Figure 6 illustrates the logical architecture, including data flows and primary components.



Figure 5: Iterative Development Cycle — plan → develop → test → review

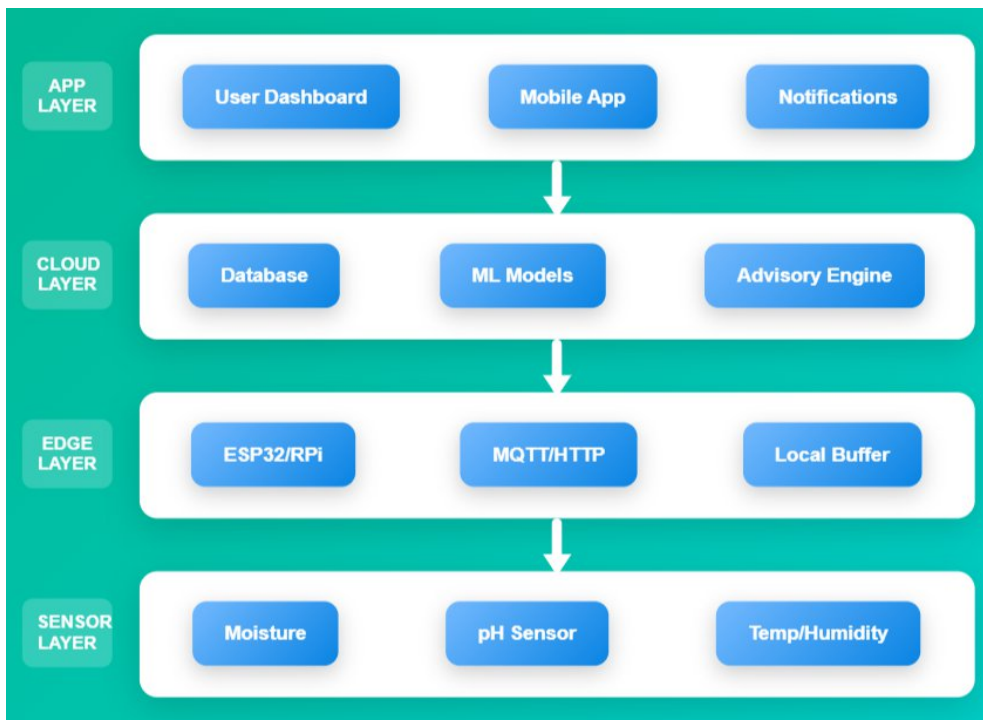


Figure 6: System Architecture — sensors → edge device (ESP32 / Raspberry Pi) → MQTT/HTTP → cloud (DB, ML, advisory engine) → user applications and notifications.

Sensor layer and edge design

Sensor selection emphasizes measurement relevance and field durability. Capacitive substrate moisture probes are preferred for wet substrates because they are less prone to corrosion and provide more stable readings than resistive alternatives. pH measurement is recognized as technically challenging in solid substrates; the protocol will compare an in-situ thin probe against periodic wet-extract sampling in initial trials to select the most reliable and maintainable procedure [2],[7]. Ambient temperature and humidity will be measured with digital sensors (e.g., SHT31). Sampling frequencies are chosen pragmatically: ambient variables at higher resolution (for example, 1-minute intervals) and moisture/pH at lower resolution (for example, 5–15 minutes), with final settings selected after calibration experiments.

At the edge, two deployment options are considered. The ESP32 microcontroller is the primary choice due to its low cost, integrated Wi-Fi, and suitability for simple telemetry; the Raspberry Pi is retained as a fallback when local processing, logging, or on-device inferencing is needed. Edge responsibilities include sensor polling, minimal preprocessing (debouncing, averaging), local buffering on network loss, and reliable forwarding using MQTT for telemetry and HTTP/REST for configuration or bulk uploads. MQTT is preferred for telemetry because of its low overhead and suitability for intermittently connected environments [1],[3].

Data collection, storage, and preprocessing

Data collection begins with controlled bench experiments designed to produce labelled sequences: scheduled watering events, controlled ambient changes, and deliberate sensor maintenance cycles. These experiments generate initial datasets that capture drying curves and pH drift required for model development. After bench validation, pilot field deployments gather operational data with event logging (user-entered watering, maintenance), enabling continual model improvement.

Data storage will record raw and calibrated readings, timestamps, device and sensor metadata, and event logs. Initial prototyping may use a managed real-time database for speed of development (e.g., Firebase), while production will favor a relational or time-series database for maintainability and analytics [3],[4]. Preprocessing follows a documented pipeline: ingestion-time plausibility checks (range and status), short-gap interpolation only when safe, resampling to uniform time steps, and feature engineering (rolling means, time-of-day features, recent watering flag). All preprocessing steps are logged so that model inputs are reproducible and audits are possible.

Predictive modelling strategy and justification

Two complementary modelling families will be evaluated: sequence neural networks (LSTM/GRU) and statistical time-series models (ARIMA/ARIMAX). This dual approach balances predictive capacity with interpretability and operational robustness.

- **Sequence models (LSTM/GRU):** These models can capture nonlinear interactions among moisture, pH, temperature, and humidity and can learn variable decay rates following watering. Input features will include recent windows of sensor readings,

event flags (watering), and temporal features (time of day). Regularization, dropout, and early stopping mitigate overfitting when data are limited; GRU is considered as a lower-complexity alternative that may converge faster on small datasets [10].

- **Statistical models (ARIMA/ARIMAX):** ARIMA models provide transparent baselines and perform reliably for short-term, autoregressive dynamics. ARIMAX allows exogenous variables (e.g., humidity, watering events) which helps model interventions that cause abrupt level changes. Model orders will be selected using standard diagnostics (ACF/PACF) and information criteria (AIC).
- **Hybrid approaches:** If neither family alone meets operational criteria, lightweight hybrids (e.g., LSTM forecasts corrected by an autoregressive residual model) will be tested, prioritizing combinational gains that preserve interpretability and manageable compute cost.

Model evaluation will use rolling origin cross-validation to mimic deployment conditions. Primary metrics are RMSE and MAE against a persistence baseline. Additionally, domain expert review of forecast plausibility will be used as a qualitative acceptance filter.

Advisory engine and decision logic

The advisory engine transforms forecasts and current measurements into actionable recommendations. Rule design follows conservative principles to avoid unnecessary alarms: an advisory is triggered only when a predicted threshold breach is expected within the advisory horizon and the breach is predicted to persist beyond a configurable time window. Rules combine absolute thresholds (e.g., moisture below a specified fraction) with trajectory-based conditions (e.g., predicted drop crossing threshold within 12 hours). Each advisory includes a recommended action, severity, and a confidence indicator derived from model uncertainty or ensemble disagreement. To reduce oscillation in advisories, fuzzy or hysteresis thresholds will be explored where appropriate, informed by prior control research in mushroom cultivation [6]. All advisories, acknowledgements, and corrective actions logged by users serve as labels for subsequent recalibration of advisory thresholds.

Implementation plan and sprint schedule

Implementation follows a sequence of focused sprints with explicit deliverables and review gates. Typical sprint sequence:

- Sprint 1: hardware selection, sensor wiring, edge firmware for logging and buffering; deliver a stable sensor logger and initial controlled dataset.
- Sprint 2: cloud ingestion pipeline and database schema, deploy a basic dashboard for live visualization; deliver end-to-end telemetry.
- Sprint 3: baseline statistical models (ARIMA/ARIMAX) and simulated LSTM prototypes using controlled data; deliver model comparison and baseline evaluation.
- Sprint 4: integrate live data with inference pipeline and surface forecasts on the dashboard; deliver integrated prediction service.
- Sprint 5: implement the advisory engine, notification pathways (SMS/Telegram), and expert review loops; deliver working advisory system.
- Sprint 6: extended field testing, metrics collection, user acceptance testing, and final refinements; deliver the evaluation report and recommended next steps.

Each sprint includes time-boxed spikes for high-risk items (for example, pH probe conditioning) and a formal review with domain experts to validate operational assumptions.

Tools, technologies, and deployment choices

Tooling choices aim to balance rapid prototyping and maintainability. Hardware: ESP32 microcontroller (primary) and Raspberry Pi (fallback) for edge tasks; capacitive moisture probes, pH probe with conditioning circuitry, SHT31/DHT22 for ambient measures. Communication: MQTT (Mosquitto) for telemetry and HTTP/REST for management and bulk operations [1],[3]. Backend: Python (FastAPI/Flask), PostgreSQL or a managed realtime DB for early prototyping [3],[4]. Machine learning: TensorFlow or PyTorch for sequence models; statsmodels for ARIMA implementations. DevOps and project workflow will use GitHub for version control and Trello or Jira for task and sprint management. These choices reflect established practices in precision agriculture projects and support quick iteration with a clear migration path to production [1],[3],[4],[9].

Evaluation, validation, and acceptance criteria

Evaluation combines quantitative and qualitative assessment. Quantitatively, forecasting models are evaluated using RMSE and MAE over rolling test windows and compared to a persistence baseline. Advisory performance is measured using precision, recall, and timeliness against logged actionable events (such as waterings recorded by users or domain experts). System reliability metrics include telemetry delivery success during network outages and mean time to recover from edge reboots. Qualitatively, domain experts review forecast plausibility and growers rate advisory clarity and usefulness in short user trials. Acceptance of the system requires consistent improvement over the persistence baseline, advisory tradeoffs agreed with cultivation experts, and stable telemetry under realistic field conditions.

Security, ethics, and governance

Security measures include authenticated APIs, encrypted transport (HTTPS and MQTT-TLS where feasible), and minimal storage of personal data. Consent is required before collecting identifiable user information; only essential contact details for notifications are retained. Advisories are explicitly framed as recommendations; automated actuation is deferred pending rigorous safety, liability, and consent protocols. Future integration of actuators will follow strict safety interlocks and testing standards, referencing established actuator designs in the literature [5],[6].

Limitations and planned extensions

Initial data scarcity limits the complexity of models that can be confidently trained; the methodology mitigates this by using statistical baselines and by collecting controlled labelled sequences early. Sensor drift and maintenance requirements necessitate scheduled calibration procedures and monitoring. Planned extensions include safe actuator integration (mist sprayers and fans with relay control), edge inferencing (TensorFlow Lite) for offline prediction, and ensemble or continual learning pipelines to manage model drift and improve long-term resilience [5],[10].

Summary and expected deliverables

This methodology maps a clear route from hardware selection to deployed advisories. Expected deliverables include: a documented sensor and edge prototype that streams validated data, a cloud ingestion and storage pipeline, a set of evaluated forecasting models with versioned artifacts, a rule-based advisory engine integrated with notification pathways, and a final evaluation report summarizing quantitative performance and expert feedback.

Project requirements

Functional requirements

Sensor data acquisition

- Description: The system must continuously collect readings from substrate moisture, pH, temperature and humidity sensors according to configured sampling rates.
- Rationale: Reliable raw data is the foundation for forecasting and advisories.
- Priority: High
- Acceptance criteria: Sensors provide timestamped data at configured intervals (e.g., humidity every 1 min, moisture/pH every 5–15 min) with $\geq 95\%$ successful samples over a 48-hour test.
- Verification: End-to-end test: deploy sensors for 48 hours, inspect DB records and telemetry logs, compute delivery success rate.

Local buffering and fault tolerance

- Description: The edge device must buffer readings locally during connectivity loss and forward buffered data when connection is restored.
- Rationale: Field networks are unreliable; buffering prevents data loss and preserves training labels.
- Priority: High
- Acceptance criteria: The edge buffers at least 24 hours of data and no records are lost after simulated 2-hour network outage.
- Verification: Simulated outage test, then confirm chronological insertion of buffered entries into cloud DB.

Secure cloud ingestion and storage

- Description: Incoming telemetry must be authenticated and stored with integrity checks and metadata.
- Rationale: Protects data integrity and user privacy.
- Priority: High
- Acceptance criteria: All ingestion endpoints require API key or token; stored records include device_id, sensor_id, timestamp, raw_value, and ingestion_status.
- Verification: API penetration test and schema inspection.

Forecast generation

- Description: The backend must produce hourly moisture forecasts (12–24h horizon) and daily pH trend estimates on a scheduled cadence (e.g., hourly).
- Rationale: Forecasts enable predictive advisories.
- Priority: High
- Acceptance criteria: Forecasts are produced within the scheduled window and attached with model_version and a simple uncertainty metric; baseline RMSE must improve over persistence baseline on validation set.
- Verification: Automated forecast job logs, model versioning records, and accuracy report against held-out data.

Advisory rule engine

- Description: The system must evaluate rules combining current readings and forecasts to generate time-bound advisories and send notifications.
- Rationale: Converts predictions into actionable guidance for growers.
- Priority: High
- Acceptance criteria: Advisories include condition, recommended action, urgency, and confidence; notifications reach configured channels (SMS/Telegram) within defined latency (e.g., <5 minutes).
- Verification: Rule unit tests, simulated forecast scenarios, and end-to-end notification delivery tests.

User feedback and logging

- Description: Users must be able to acknowledge advisories and log corrective actions; logs must be stored for retraining.
- Rationale: Enables supervised recalibration and improves trust.
- Priority: Medium
- Acceptance criteria: Dashboard provides an acknowledge/action button; acknowledgements are timestamped and linked to the originating advisory.
- Verification: UI acceptance test and DB verification of acknowledgement records.

Non-functional requirements

Availability and resilience

- Target: System availability $\geq 99\%$ (excluding planned maintenance).
- Measurement: Uptime monitoring and incident logs.
- Verification: Monitor for one deployment month and compute availability.

Data durability

- Target: No loss of persisted data; backups retained daily with 30-day retention.
- Verification: Backup restore drill.

Latency and timeliness

- Target: Telemetry ingestion within 30s of publish (under typical network conditions); advisory notifications delivered within 5 minutes after rule trigger.
- Verification: Performance tests under representative network conditions.

Scalability

- Target: Cloud design should support scaling to dozens of grow rooms without architecture changes. Systems must scale horizontally for ingestion and model serving.
- Verification: Load test with simulated device fleet.

Security and privacy

- Requirements: TLS for transport, token/API key authentication, role-based access to UI, minimal personal data storage with consent records.
- Verification: Security checklist and basic vulnerability scan.

Maintainability and observability

- Requirements: Modular code, versioned artifacts, logs and metrics for telemetry, model performance, and system health.
- Verification: Code review, CI tests, and observability dashboard configured.

User requirements

Clear and actionable advisories

- Users want simple messages telling them what to do, why, and by when (e.g., “Moisture predicted to drop below 50% within 8 hours — irrigate within 6 hours”).
- Acceptance: At least 80% of trial participants rate advisories as “clear and actionable” in usability survey.
- Verification: Short user trials and survey.

Low-cost and easy-to-maintain hardware

- Users prefer affordable sensors and simple maintenance (replaceable probes, clear calibration steps).
- Acceptance: Total hardware cost within target budget; maintenance guide with ≤ 10 steps.
- Verification: Bill of materials, maintenance checklist, and user feedback.

Minimal technical overhead

- Users expect minimal setup effort and a responsive support path (help documentation or short support calls).
- Acceptance: Average setup time under 2 hours for a single grow room with provided instructions.
- Verification: Time-to-deploy trials with novice users.

Transparent recommendations and confidence

- Users want to understand why advice is given and how certain the system is.
- Acceptance: Each advisory displays a short rationale and confidence level.
- Verification: UI inspection and user feedback.

System requirements (hardware, software, interfaces, constraints)

Edge hardware

- Minimum: ESP32 or equivalent microcontroller with ADC and Wi-Fi; optional Raspberry Pi for local DB/processing.
- Constraints: Must support local buffering (≥ 24 h) and OTA firmware updates.

Sensors and signal conditioning

- Capacitive moisture probe, pH probe with conditioner, digital temp/humidity (SHT31/DHT22).
- Calibration and replacement procedures documented.

Communication interface

- MQTT broker for telemetry; HTTP/REST for management APIs.
- Support for intermittent connectivity and buffering

Cloud infrastructure

- Backend service (Python/Node) with a relational/time-series DB (Postgres/Influx) or managed realtime DB for prototyping.
- Model serving endpoint and job scheduler for periodic retraining/inference.

Security & integration

- TLS, API tokens, user authentication; optional integration points for SMS gateway and instant messaging APIs.

Logging, monitoring and backups

- Centralized logs, metrics, and daily backups with documented restore procedure.

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Appendices

The project timeline is shown in Figure 7.

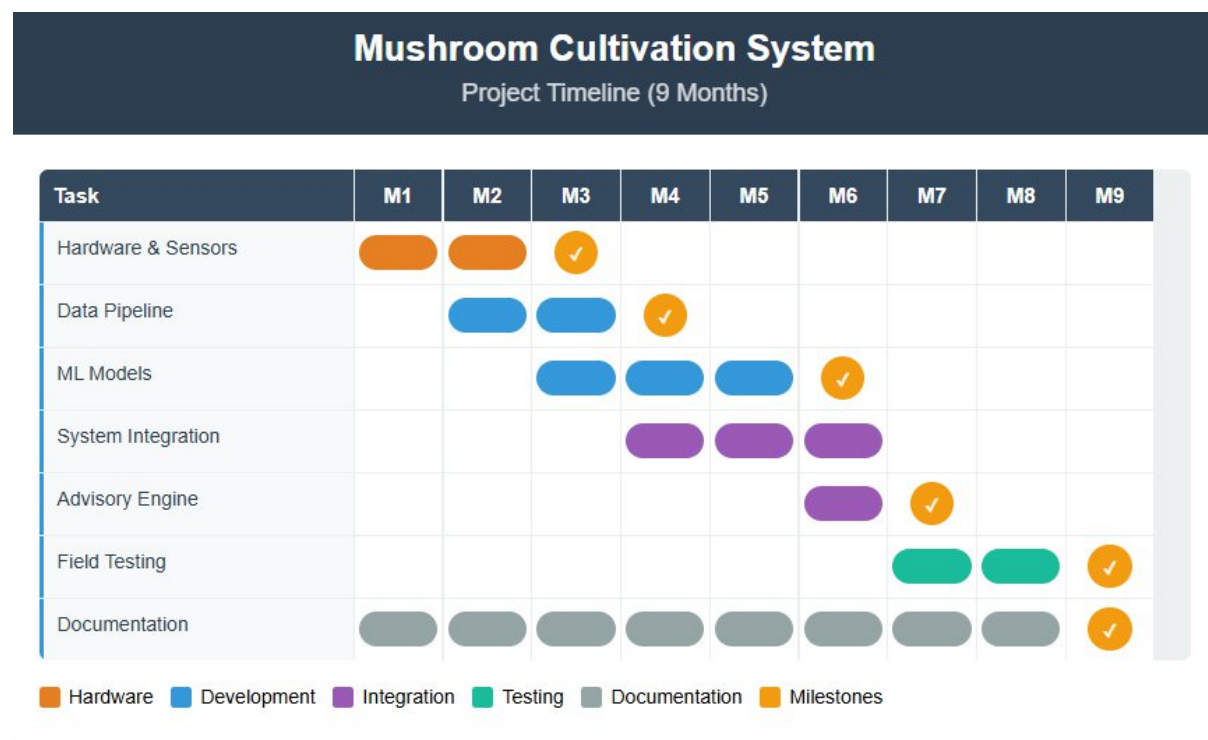


Figure 7: 18-Months sprint timeline (visual) and brief legend for colors, month units, and assumptions

6. Work Breakdown Structure

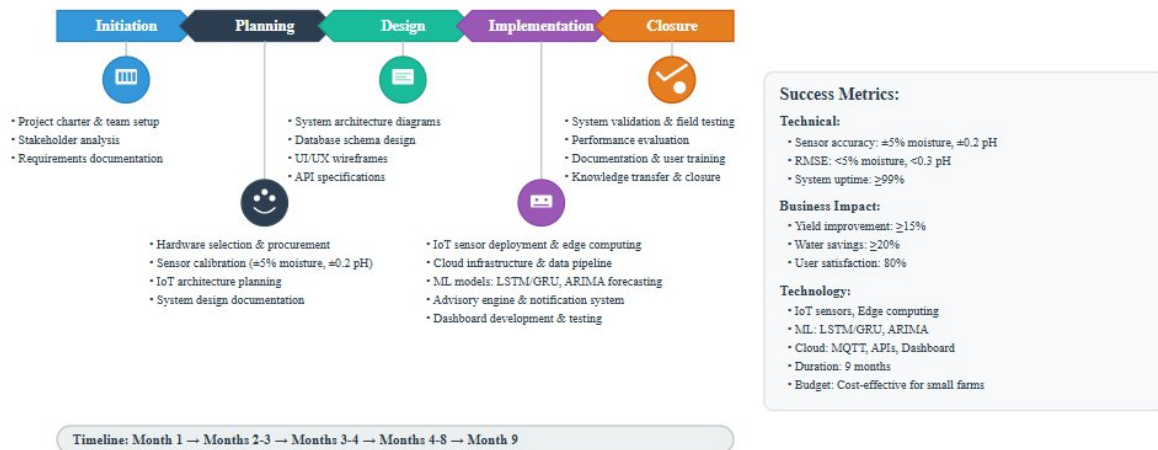


Figure 7: 18-Months sprint timeline (visual) and brief legend for colors, month units, and assumptions